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The title of your publication should effectively capture the larger breadth of inquiry in a manner that is not ambiguous to the reader. Titles should be concise, informative, and free of grammatical errors.

XXX (Principal Author), XXX (Bench-mate 1)

COURSE CODE: XXX INSTRUCTOR NAME: XXX

Abstract: The goal of the abstract is to concisely communicate a summary of your experiment. It should give the reader an idea of the objective, procedure, and significance of your experiment. Follow your instructor's guidelines for length; however, most are limited to one paragraph of 100-200 words. A good method of summarizing your report is to have one sentence dedicated to each of these sections: introduction, methods and materials, results, and discussion.

Keywords: Include any relevant keywords that best describe the nature of your publication.

Introduction/Theory

The *Introduction* traces the background concerning your particular experiment while acknowledging the existing gaps in knowledge. This will provide the context of your experiment – describe how the present study will try to address those gaps. Ensure that the objective and larger purpose of the report is clearly stated in this section. This is a good place to include in-text citations of secondary sources that you may find in your research. State your hypotheses about the outcome of the experiment and any predictions based on it near the end of this section. When constructing your hypotheses, your educated guess as to the outcome of the experiment should offer a decisive prediction without restating a given fact.

Materials and Methods/Procedure

The purpose of the *Methods and Materials* section is to give enough information to allow someone else to replicate the experiment. This section consists of either a list or paragraph of all the apparatus used and steps taken. Be comprehensive and precise in your descriptions. You may choose to include a labelled schematic of your experimental setup if needed.

Data Collection and Quantitative Analysis

This section is where you will describe any calculations or special formulae that were used to assemble information relevant to the experiment. Ensure that the calculations are simple enough to understand by the layperson. All calculations must be thoroughly expanded upon and rules of significant digits must be followed.

Results

The *Results* section presents your actual findings without analyzing or interpreting them. Describe your results in paragraph form and include relevant tables, figures, and graphs. Calculations can be written in full once, displayed in a table for all others, or may be referenced from the *Data Collection and Quantitative Analysis* portion of your formal lab report. Reference your figures and tables with descriptive captions.

Discussion

The purpose of the *Discussion* is to state the implications of the results of the experiment. You should refer back to your hypothesis and predictions and state if they were proven. Interpret your data from the results section and include references to other research for comparison. Try to answer the following questions:

- Did you get the expected results? What were the expected results? What could account for discrepancies between the actual and theoretical values?
- Are your results supported by similar experiments in the larger body of academic literature?
- Does the experimental design have any limitations? What are its strengths? What sources of error or difficulty did you encounter during the experiment?
- How might these have affected your results and how can they be avoided in the future?

Wrap-up this section by providing the reader with a conclusion for stimulating further thought on the topic and suggest areas for future research.

Conclusion

This section can be very short in most high school lab reports and well into undergraduate education. The purpose of the *Conclusion* is to simply state what you know now for sure, as a result of the lab. Ensure that you answer the procedure statements presented in the *Introduction*. Comment on your hypothesis. If the hypothesis is incorrect, state a new one. The conclusion acts as a larger summary of the experiment and the results.

References

Always make sure to cite your sources. The precise format of citation may vary from discipline to discipline, so refer to specific instructions. For example, APA citations are commonly used in physics while biology uses CSE style citations. In-text citations should always be used where appropriate.

Appendix

This section is not always required. *Appendices* are for information that would otherwise be too detailed to include in either the Methods and Materials or the Results sections. If you include appendices, they must be directly referred to in your lab report

Acknowledgements

This is the section where you may comment on external sources of success or motivation. Those who have contributed to the success of this lab report may be honorably mentioned here.